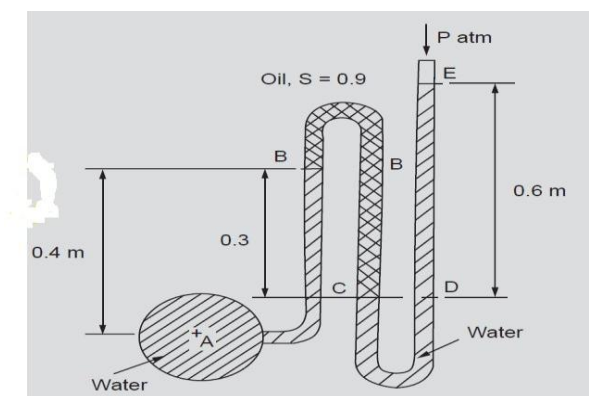


Subject: Fluid Mechanics and Machinery (ME205n)

UNIT 1

ASSIGNMENT 1

1. Define viscosity and explain its physical origin in both liquids and gases. Differentiate clearly between dynamic and kinematic viscosity, and derive its dimensions in the [MLT] system.
2. Discuss the various types of fluids based on their rheological behavior. Explain Newtonian, Bingham plastic, pseudoplastic, and dilatant fluids with suitable examples and diagrams.
3. A rectangular plate of size 50 cm $\times$ 50 cm and weighing 150 N slides down an inclined plane at 30° to the horizontal. A layer of oil 1.5 mm thick is maintained between the plate and the plane. If the plate slides at a constant velocity of 0.5 m/s, find the dynamic viscosity of the oil in Poise. If the specific gravity of the oil is 0.9, calculate the kinematic viscosity in Stokes.
4. Explain the terms: Absolute pressure, Gauge pressure, Vacuum pressure, and Atmospheric pressure. Show these relationships on a standard pressure-datum diagram. A multiple U-tube manometer is fitted to a pipe with centre at A as shown in Fig. 2. Determine the pressure at A.



5. Derive an expression for the total hydrostatic force and the center of pressure for a vertical plane surface submerged in a liquid. Why is the center of pressure always below the centroid of the area?
6. State Archimedes' Principle and derive the expression for the buoyant force. Explain the concept of the center of buoyancy. Explain the conditions for the stable, unstable, and neutral equilibrium of a completely submerged body. Draw the relative positions of G and B for each case and explain the nature of the couple produced when the body is tilted.

7. Explain the relationship between the metacentric height and the period of rolling of a ship. Why is a very large metacentric height undesirable for passenger ships?
8. Define the following fluid flow terms: (a) steady and unsteady flow, (b) Uniform and non-uniform, (c) Laminar and turbulent flow, (d) Rotational and Irrotational flow, (e) Stream line, (f) Path line, (g) Streak line

Provide brief explanations and, where appropriate, illustrate the concepts with suitable examples or diagrams.

9. Derive the general form of the continuity equation (conservation of mass) for fluid flow. Clearly state all assumptions made during the derivation and simplify the equation for one-dimensional steady incompressible flow.
10. The three-dimensional velocity field of a fluid flow is given by:

$$\vec{V} = (2xt + y^2) \hat{i} + (3yz + t^2) \hat{j} + (xz + 4t) \hat{k}$$

Determine the following at the point (x,y,z) = (1,2,1), and time t=2s

- (a) Local (temporal) acceleration
- (b) Convective acceleration
- (c) Total acceleration vector
- (d) Magnitude of the total acceleration